

Seonghyeon Lim

KAIST, Daejeon, South Korea
Ph.D. Candidate in Electrical Engineering

+82-10-8488-1838
shlim@kaist.ac.kr
Google Scholar

RESEARCH INTERESTS

- Robot Learning (especially online/offline reinforcement learning)
- High-DoF legged robot manipulation (e.g., humanoid, Quadruped manipulator, and quadruped robot)

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Ph.D. Candidate in Electrical Engineering — Sep. 2026 – Present
Advisors: Hyun Myung and Kimin Lee

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
M.S. in Electrical Engineering — Sep. 2024 – Aug. 2026
Advisor: Hyun Myung

Sogang University — Seoul, South Korea
B.S. in Mechanical Engineering — Mar. 2018 – Aug. 2024
GPA: 4.02/4.50

PUBLICATIONS

[1] Seonghyeon Lim, K. Lee, and H. Myung. Adaptive Value Anchoring for Long-Horizon Stitching in Offline Reinforcement Learning. International Conference on Learning Representations **(ICLR)**, 2027. Submitted.

[2] Seonghyeon Lim, H. Lee, M. Kim, K. Lee, and H. Myung. MAPO: Mixture of Experts with Alternating Policy Optimization for Universal Humanoid Motion Tracking. Conference on Robot Learning **(CoRL)**, 2026. Accepted.

[3] Seonghyeon Lim, H. Lee, S. Lee, I. M. A. Nahrendra, and H. Myung. OpenHEART: Opening Heterogeneous Articulated Objects with a Legged Manipulator. IEEE International Conference on Robotics and Automation **(ICRA)**, 2026. **Oral presentation.** **[Project]**

[4] S. Kwon*, Seonghyeon Lim*, H. Lee, and H. Myung. Reinforcement Learning-based Inverse Kinematics Compensation for Robust Manipulation under Payload Variation and Singularity. International Conference on Control, Automation and Systems **(ICCAS)**, 2025.

[5] J. Shim, Seonghyeon Lim, and C. Nam. ST-CBS: Spatio-Temporal Conflict Based Search in Continuous Space for Multi-Agent Pathfinding. ACM/SIGAPP Symposium on Applied Computing **(SAC)**, 2025.

[6] Seonghyeon Lim*, H. Lee*, S. Lee, and H. Myung. Learning Robust Whole-Body Control for Legged Manipulation via Learned Terrain Estimation. International Conference on Control, Automation and Systems **(ICCAS)**, 2024.

[7] J. Hamm*, Seonghyeon Lim*, J. Park, J. Kang, I. Lee, Y. Lee, J. Kang, Y. Jo, J. Lee, S. Lee, A. Brilian, S. Lee, and K. Shin. A Modular Robotic Platform for Automated and Remote Operation of Chemical and Biological Experiments in University Laboratories. Advanced Intelligent Systems, 2024. **Highlighted on the inside cover.** **[Project]**

[8] J. Jeong, I. Kim, Y. Choi, Seonghyeon Lim, S. Kim, H. Kang, D. Shah, R. Baines, J. Booth, R. Bottiglio, and S. Kim. Spikebot: a Multigait Tensegrity Robot with Linearly Extending Struts. Soft Robotics, 2023.

* Equal contribution.

HONORS & AWARDS

Second Place May 2025
Quadruped Robot Challenges, ICRA 2025 Atlanta, GA, USA
Organizer: IEEE Robotics and Automation Society
Contributions:

- Developed a whole-body controller for a legged manipulator.
- Developed a framework for autonomous mission execution.
- Developed sensor housing and electrical components.

First Prize in Undergraduate Student Research Dec. 2022
Department of Mechanical Engineering Academic Seminar Seoul, South Korea
Organizer: Department of Mechanical Engineering, Sogang University

Gold Medal Nov. 2022
International Genetically Engineered Machine Competition (iGEM 2022) Paris, France
Organizer: iGEM Foundation
Nominations: Best Software, Best Hardware, and Best Presentation.
Project: Autonomous biological experiment system using a robotic arm.
Contributions:

- Developed rule-based task plans.
- Developed motion primitives, including pipetting and opening an incubator.
- Designed auxiliary devices for laboratory automation.

Best Poster Presentation Award Aug. 2022
International Conference on Energy, Aquatech and Sustainability (ICEAS 2022) Ansan, South Korea
Organizer: Academic Society for Appropriate Technology

ROBOT PLATFORM EXPERIENCE

Humanoids	G1 (29 DoF and 23 DoF)
Quadrupeds	Go2, A2
Manipulators	ViperX, Nero
Quadruped Manipulators	Go2 + ViperX, A2 + Nero